

[Title TBD]

DSAN 5450 — Final Project

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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import StratifiedKFold, cross_val_predict
from sklearn.metrics import roc_auc_score, roc_curve, accuracy_score

np.random.seed(5450)
```

Introduction

The median voter theorem, first postulated by [inserrt], is a concept in political science arguing that in democratic elections, candidates tailor their campaigns and policies to the pivotal “swing” voter who gives them the bare plurality of the vote. Much abstraction of political opinions is required to conceptualize who the median voter is, oftentimes by placing all voters’ ideologies onto a single line and finding who lies in the middle. Political scientists have used the median voter as an abstraction to represent the moderating tendencies of a two-party democracy [citation needed], where both parties will pivot to the center in an attempt to gain a marginal advantage over their rival.

The median voter theorem plays out in real time with the 538 members of the electoral college. Unless you live in one of a few swing states, it is highly unlikely for a presidential candidate to show up in your town to gladhand or tailor their policies to match your parochial interests (note the precipitous decline in ethanol white papers as Iowa has fallen off the political battleground). The causation of candidates’ tunnel vision on the swing states is clear: a Democratic presidential candidate knows with great certainty that they have no shot of winning in South Dakota and will not waste their precious reserves attempting to turn out their base or persuade

swing voters in Sioux Falls. A Republican is not spending on broadcast in Los Angeles hoping to erode the massive Democratic lean of California. States as a whole are largely disenfranchised from the presidential election solely based on the certainty that they will vote one way or another.

Within swing states, however, candidates are forced to actually campaign to turn out their supporters and convince fence-sitters to join their camp. Billions of dollars are spent on elections in swing states as campaigns blanket the airwaves with ads, streets with canvassers, and inboxes with mailers. Given the narrow margins in pivotal states like Pennsylvania, Michigan, and Georgia, candidates are more than willing to expend massive resources to communicate and engage these states electorates.

But what if a campaign could actually tailor their campaign to the apocryphal median voter (or a finite subset of the electorate most likely to be pivotal)? What if they could determine their address, their media consumption habits, and what messages could persuade them most effectively? What if swing voters were people who could be determined with certainty rather than pundits' crude characterizations? The modern political data industry makes that level of microtargeting possible. The vast swathes of consumer data available for purchase allow for the construction of intricate models of voter behavior, enabling campaigns to pick and choose which members of the electorate are worth engaging the in the democratic process and which people are a lost cause, either because modeling has rendered them hostile to a candidate's positions or unlikely to actually vote. Further than restricting spending to a select number of states likely to be competitive in an election, campaigns can suss out a subset of actual people within a state who are likely to be the proverbial median voter, as the massive data at their disposal credited the uncertainty that would otherwise force them to cast a wide net and engage these entire electorate in the democratic process.

Data and Methods

To simulate the effects of growing datasets on vote choice certainty, I build predictive models of partisanship and turnout based on responses from the ANES/CCES. Voter files contain only basic information about voters, typically their age, gender, name, and address. Some states also track party registration or keep records of voters' race to comply with the Voting Rights Act. The former set of variables may give some indication of partisanship, and the latter gets them much of the way to constructing a full model of a voter's behavior.

[build out model here. One with only age and gender as variable and another incorporating race and party registration].

```

cols = ['birthyr', 'gender4', 'race', 'educ', 'faminc_new', 'marstat', 'employ',
        'ownhome', 'investor', 'pid3']
raw = pd.read_csv('data/CCES22_Common_OUTPUT_vv_topost.csv',
                  usecols=cols, low_memory=False)

df = raw.copy()
df.loc[df['faminc_new'] == 97, 'faminc_new'] = np.nan
df['age'] = 2022 - df['birthyr']
df['gender_bin'] = df['gender4'].apply(lambda x: 1 if x == 1 else 2)
df = df[df['pid3'].isin([1, 2]).copy()]
df['target'] = (df['pid3'] == 1).astype(int) # 1 = Democrat, 0 = Republican

for col in ['faminc_new', 'marstat', 'employ', 'ownhome', 'investor']:
    df[col] = df[col].fillna(df[col].median())

F1 = ['age', 'gender_bin']
F2 = ['age', 'gender_bin', 'race']
F3 = ['age', 'gender_bin', 'race', 'educ', 'faminc_new',
      'marstat', 'employ', 'ownhome', 'investor']

y = df['target'].values

```

```

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
rf = RandomForestClassifier(n_estimators=200, min_samples_leaf=20,
                           class_weight='balanced', random_state=42, n_jobs=-1)

probas = {}
for name, feats in [('M1', F1), ('M2', F2), ('M3', F3)]:
    probas[name] = cross_val_predict(rf, df[feats].values, y,
                                     cv=cv, method='predict_proba')[:,1]

    auc = roc_auc_score(y, probas[name])
    acc = accuracy_score(y, probas[name] >= 0.5)
    print(f"{name}: AUC={auc:.3f} Accuracy={acc:.3f}")

```

Already, these barebones demographic models give campaigns a clear view of who is likely to be a swing voter and who is nearly certainly a Democrat or a Republican. How do they go the extra mile? Consumer data collected by companies like Experian, Acxiom, etc. contain thousands of variables measuring or inferring traits like education, income, marital status, and consumption preferences. [talk about increasing educational polarization]. Incorporating these variables enables a campaign to predict with a voters' preferences with near certainty. Even this example is limited, relying on a few extra variables on the ANES/CCES which do not

come close to touching the vast depths of consumer information available to savvy campaigns. [see if I can drudge up anything about Romney dog example].

Results

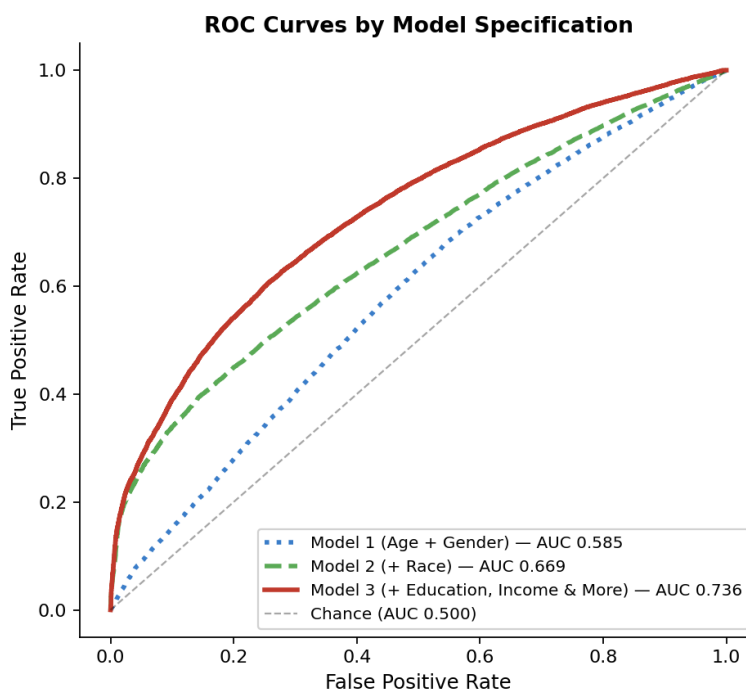


Figure 1: ROC curves for all three model specifications.

Take an example of a 1,000 person electorate. If a campaign has a budget of a million dollars, how would they allocate to maximize the chance of talking to the median voter. To vastly oversimplify this decision, assume that they spend proportional to the uncertainty associated with a voter. If voters are scored from 0-100 on their likelihood of voting for a candidate, we might assume the campaign spending would follow a normal distribution centered at 50, with most of the spending focused on the people a campaign does not have a good handle on. At the edges of this distribution, spending falls precipitously as campaigns avoid wasting resources on people they assume are already supporting them or would never vote for them.

FUTURE GOAL: Enter your own demographic information here (or use your internal biases to construct a swing voter or hard partisan) and watch how a campaign chooses to allocate resources to you based on their increasingly certain assumptions of your preferences.

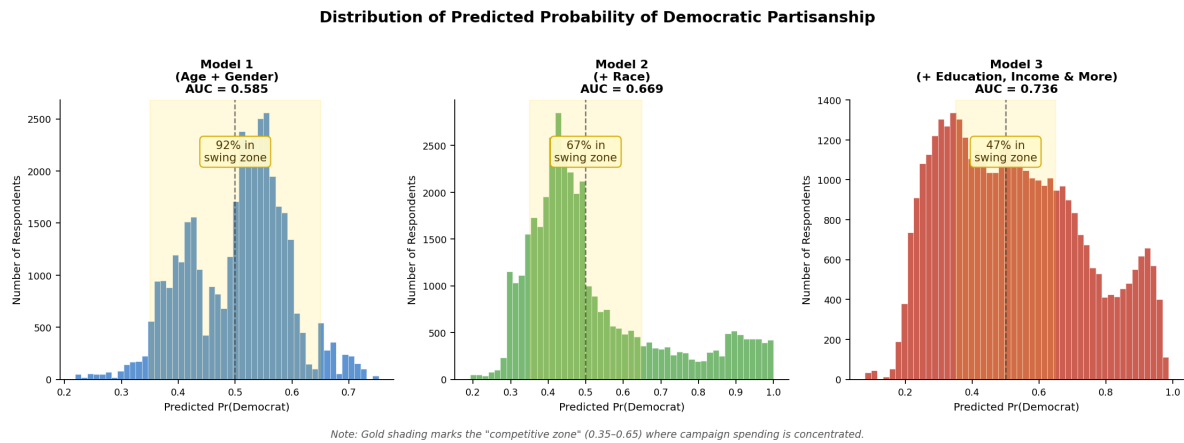


Figure 2: Distribution of predicted probability of Democratic identification across three model specifications. Gold shading marks the “competitive zone” (0.35–0.65).

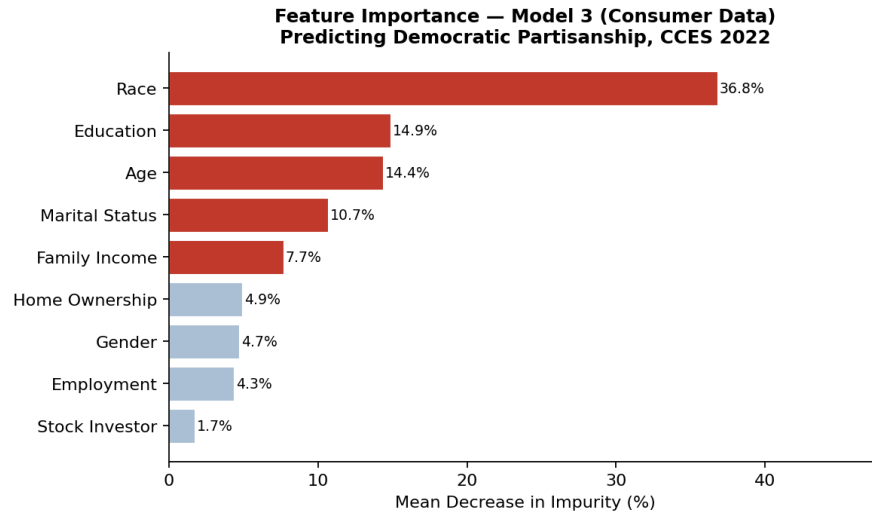


Figure 3: Feature importance for Model 3 (consumer data).

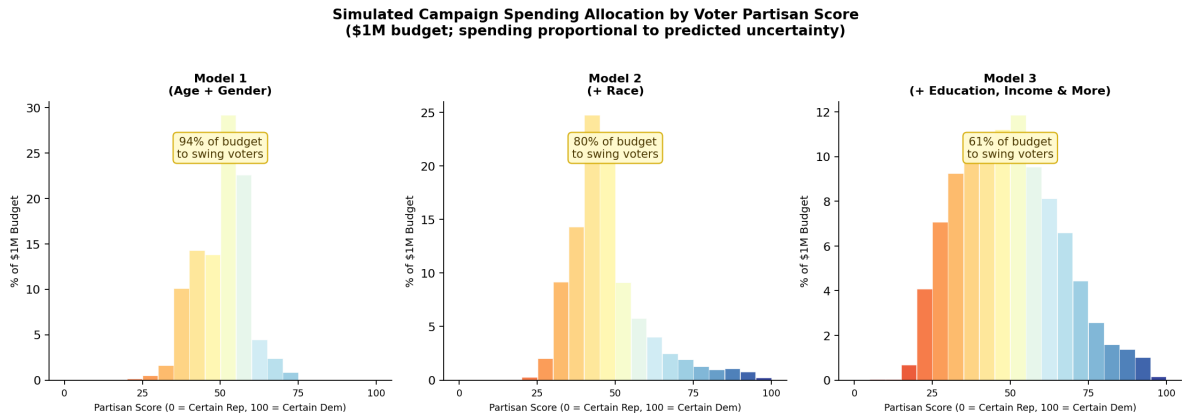


Figure 4: Simulated campaign spending allocation across voter partisan scores under each model specification.

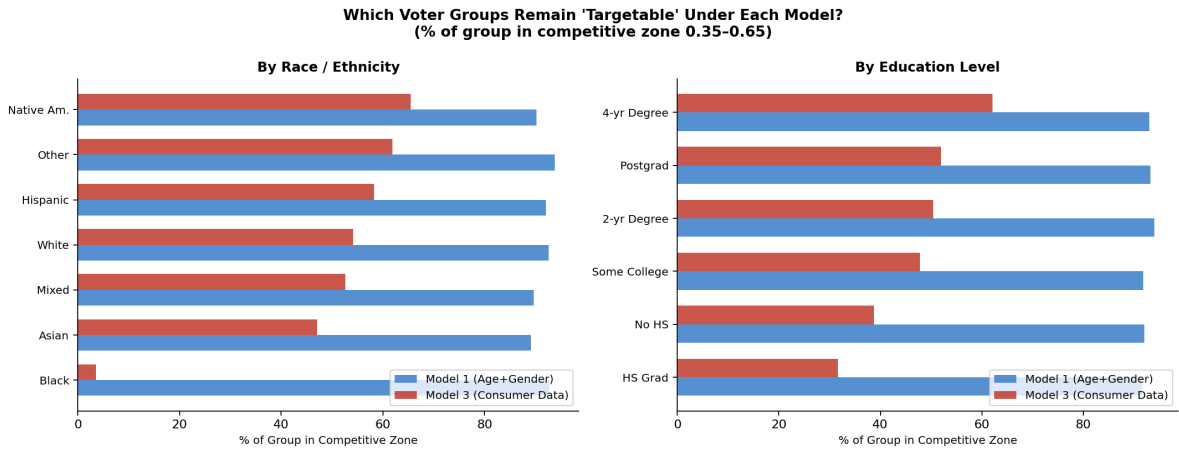


Figure 5: Percentage of each demographic group remaining in the competitive zone under the voterfile-only vs. consumer data models.

Policy Recommendation

What is a way out of this narrowing of the electorate? The siloing of consumer data from political campaigns would be a good start, limiting the ability of campaigns to pick and choose which members of the public to include in their communications.

References